

LIFESTYLE DYNAMICS FRAMEWORK

A Data-Driven Analysis of Sleep, Stress, Physical Activity and Health
Outcomes using Statistical Analysis & Machine Learning

PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the Course **Data Science Summer Training Course**

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Project Duration

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Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Statistical Analysis
- K-Means Clustering

Key Areas Covered

Data Cleaning & Preprocessing
Exploratory Data Analysis
Statistical Analysis
Correlation Analysis
ANOVA Test
Chi-Square Test
Lifestyle Segmentation using K-Means
Personalized Recommendations

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Abstract

In today's fast-paced world, maintaining a healthy lifestyle has become increasingly challenging due to factors such as stress, irregular sleep schedules, lack of physical activity, and unhealthy habits. These factors significantly influence an individual's physical health, mental well-being, and overall quality of life. Among them, sleep health plays a crucial role in maintaining productivity, emotional stability, and long-term wellness. Therefore, understanding the relationship between sleep-related factors and lifestyle habits is essential for promoting healthier living.

The objective of this project is to analyze the impact of sleep duration, sleep quality, stress level, physical activity, BMI category, occupation, and other health-related factors on overall lifestyle and well-being. The Sleep Health and Lifestyle Dataset containing 374 records and 13 features was used for this study. Data preprocessing techniques such as missing value handling, duplicate verification, and data validation were performed to ensure data quality.

Exploratory Data Analysis (EDA) was conducted to identify trends, patterns, and relationships among variables. Statistical techniques including Correlation Analysis, ANOVA Test, and Chi-Square Test were applied to validate significant relationships between different lifestyle factors. In addition, K-Means Clustering was used to segment individuals into distinct lifestyle groups based on sleep, stress, physical activity, heart rate, and daily steps.

The results revealed a strong negative relationship between sleep quality and stress level, indicating that higher stress is associated with poorer sleep quality. The analysis also showed that occupation significantly affects sleep quality, while BMI category is strongly associated with sleep disorders. Using K-Means Clustering, three meaningful lifestyle groups were identified: Optimal Lifestyle, Active but Stressed, and High-Risk Lifestyle.

Based on these findings, personalized recommendations were generated to help individuals improve sleep quality, reduce stress, and adopt healthier lifestyle habits. This project demonstrates how statistical analysis and machine learning techniques can be used to uncover meaningful health patterns and support data-driven decision-making for improved well-being.

Keywords: Sleep Health, Lifestyle Analysis, Statistical Analysis, K-Means Clustering, Machine Learning, Stress Level, Physical Activity, Health Analytics.

Chapter 1: Introduction

1.1 Background

Sleep plays a vital role in maintaining physical and mental health. Poor sleep quality, high stress levels, low physical activity, and unhealthy lifestyle habits contribute to various health disorders and reduced quality of life.

With the increasing availability of health-related data, data analytics and machine learning techniques can be used to identify lifestyle patterns and provide personalized recommendations for improving health outcomes.

1.2 Problem Statement

Lifestyle factors such as:

- Sleep Duration
- Quality of Sleep
- Stress Level
- Physical Activity Level
- BMI Category
- Occupation

significantly influence overall health and well-being.

Research Question:

How do sleep, stress, physical activity, BMI, and occupation affect human lifestyle and health outcomes?

1.3 Objectives

The primary objectives of this project are:

1. Analyze the relationship between sleep and lifestyle factors.
2. Identify significant health indicators affecting sleep quality.
3. Perform statistical testing to validate relationships.
4. Segment individuals into meaningful lifestyle groups.
5. Generate personalized recommendations based on lifestyle patterns.

Chapter 2: Dataset Description

2.1 Dataset Information

Dataset Name: Sleep Health and Lifestyle Dataset

Total Records: 374

Total Features: 13

2.2 Features Used

Sleep Features

- Sleep Duration
- Quality of Sleep
- Sleep Disorder

Health Features

- BMI Category
- Heart Rate
- Blood Pressure

Lifestyle Features

- Physical Activity Level
- Daily Steps
- Stress Level

Personal Features

- Age
- Gender
- Occupation

Chapter 3: Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed to ensure that the dataset was accurate, consistent, and suitable for analysis and machine learning. These steps helped improve data quality and reliability.

3.1 Data Loading

The dataset was imported using the Pandas library and examined using functions such as `head()`, `info()`, and `describe()` to understand its structure, features, and data types.

3.2 Missing Value Analysis

All columns were checked for missing values.

Observation

- Most columns contained no missing values.
- The **Sleep Disorder** column contained missing values.

3.3 Missing Value Handling

Missing values in the **Sleep Disorder** column were replaced with "**None**" to represent individuals without diagnosed sleep disorders. This ensured data consistency and prevented information loss.

3.4 Duplicate Check

The dataset was checked for duplicate records.

Result

- No significant duplicate records were found.
- The dataset was considered clean and reliable for analysis.

3.5 Data Type Verification

Numerical and categorical variables were verified and stored using appropriate data types to ensure accurate statistical analysis and machine learning results.

Chapter 4: Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset, identify trends, discover hidden patterns, and analyze the relationships between various sleep, health, and lifestyle factors. The objective of EDA was to gain meaningful insights before applying statistical analysis and machine learning techniques.

Analysis Performed

Sleep Analysis

The following sleep-related variables were analyzed:

- Sleep Duration
- Quality of Sleep
- Sleep Disorder

Distribution plots and summary statistics were used to understand sleeping patterns among individuals. The analysis revealed noticeable variations in sleep duration and sleep quality across different individuals.

Health Analysis

Health-related factors analyzed included:

- BMI Category
- Heart Rate
- Blood Pressure

The distribution of BMI categories showed the presence of normal, overweight, and obese individuals. These variables were further explored to understand their impact on sleep quality and sleep disorders.

Lifestyle Analysis

Lifestyle factors examined during EDA included:

- Physical Activity Level
- Daily Steps
- Stress Level

The analysis showed that individuals with higher physical activity levels generally demonstrated better lifestyle patterns. Stress level emerged as one of the most influential factors affecting sleep quality and overall well-being.

Demographic Analysis

Demographic variables such as:

- Age
- Gender
- Occupation

were analyzed to identify differences in sleep behavior and health outcomes among different population groups.

Visualizations Used

Several visualization techniques were utilized to support the analysis:

- Histograms
- Count Plots
- Distribution Plots
- Correlation Heatmaps

These visualizations helped identify trends, distributions, and relationships between different variables.

Conclusion of EDA

The Exploratory Data Analysis provided valuable insights into the relationship between sleep, stress, physical activity, BMI, and overall lifestyle health. The findings indicated that sleep quality and stress level are among the most important factors influencing well-being. These insights formed the foundation for the subsequent statistical analysis, hypothesis testing, and K-Means clustering performed in the project.

Chapter 5: Statistical Analysis

5.1 Correlation Analysis

Correlation analysis was performed to examine relationships between numerical variables.

Key Findings

Variables	Correlation
Sleep Quality vs Stress Level	-0.899
Sleep Duration vs Stress Level	-0.811
Sleep Duration vs Heart Rate	-0.516
Sleep Quality vs Physical Activity	0.193

5.2 ANOVA Test

Purpose:

To determine whether sleep quality differs significantly across occupations.

Result:

ANOVA test showed statistically significant differences in sleep quality among occupations.

Conclusion:

Occupation significantly affects sleep quality.

5.3 Chi-Square Test

Purpose:

To evaluate the association between BMI Category and Sleep Disorder.

Result:

Chi-Square test showed a significant relationship.

Conclusion:

BMI Category is significantly associated with sleep disorders.

Chapter 6: Lifestyle Segmentation using K-Means Clustering

6.1 Why K-Means?

K-Means clustering is an unsupervised machine learning algorithm used to group individuals with similar characteristics.

6.2 Features Used

- Sleep Duration
- Quality of Sleep
- Physical Activity Level
- Stress Level
- Heart Rate
- Daily Steps

6.3 Optimal Number of Clusters

The Elbow Method was used to determine the optimal number of clusters.

Result:

Optimal Number of Clusters = 3

6.4 Cluster Distribution

Cluster	Individuals	Percentage
Cluster 0	239	63.9%
Cluster 1	34	9.1%
Cluster 2	101	27.0%

Chapter 7: Lifestyle Cluster Profiles

Cluster 0 – Optimal Lifestyle

Individuals: 239 (63.9%)

Characteristics:

- High sleep quality
- Healthy activity level
- Low stress level
- Balanced lifestyle

Cluster 1 – Active but Stressed

Individuals: 34 (9.1%)

Characteristics:

- High physical activity
- High daily steps
- Elevated stress level

Cluster 2 – High Risk Lifestyle

Individuals: 101 (27.0%)

Characteristics:

- Poor sleep quality
- Lower activity level
- High stress level
- Unhealthy lifestyle pattern

Chapter 8: Personalized Recommendations

Optimal Lifestyle

Recommendations:

- Maintain current sleep habits
- Continue regular exercise
- Maintain low stress levels

Active but Stressed

Recommendations:

- Reduce stress levels
- Practice meditation
- Improve recovery and sleep quality

High Risk Lifestyle

Recommendations:

- Increase physical activity
- Improve sleep duration
- Reduce stress through healthy habits

Chapter 9: Results and Findings

The analysis revealed the following:

1. Sleep Quality and Stress Level showed the strongest negative relationship (-0.899).
2. Sleep Duration is strongly associated with lower stress levels.
3. Physical Activity positively influences sleep quality.
4. Occupation significantly affects sleep quality.
5. BMI Category is significantly associated with sleep disorders.
6. Three meaningful lifestyle groups were identified using K-Means clustering.

Chapter 10: Conclusion

This project successfully analyzed the relationship between sleep health and lifestyle factors using statistical analysis and machine learning techniques.

The findings indicate that sleep quality, stress level, physical activity, BMI, and occupation significantly influence overall lifestyle and health outcomes. K-Means clustering effectively segmented individuals into meaningful lifestyle groups, enabling personalized recommendations.

The study demonstrates the importance of healthy sleep habits, regular physical activity, and stress management in improving overall well-being.

Future Scope

- Develop a real-time lifestyle monitoring system.
- Integrate wearable health devices.
- Implement AI-powered recommendation systems.
- Build predictive models for sleep disorder detection.
- Create personalized health dashboards.

Final Takeaway

Better Sleep + Lower Stress + Regular Physical Activity = Healthier Lifestyle & Improved Well-Being